

2021

AP[®]

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AP[®] Physics C: Electricity and Magnetism

Free-Response Questions



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ADVANCED PLACEMENT PHYSICS C TABLE OF INFORMATION

CONSTANTS AND CONVERSION FACTORS	
Proton mass, $m_p = 1.67 \times 10^{-27} \text{ kg}$	Electron charge magnitude, $e = 1.60 \times 10^{-19} \text{ C}$
Neutron mass, $m_n = 1.67 \times 10^{-27} \text{ kg}$	1 electron volt, $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$
Electron mass, $m_e = 9.11 \times 10^{-31} \text{ kg}$	Speed of light, $c = 3.00 \times 10^8 \text{ m/s}$
Avogadro's number, $N_0 = 6.02 \times 10^{23} \text{ mol}^{-1}$	Universal gravitational constant, $G = 6.67 \times 10^{-11} (\text{N}\cdot\text{m}^2)/\text{kg}^2$
Universal gas constant, $R = 8.31 \text{ J}/(\text{mol}\cdot\text{K})$	Acceleration due to gravity at Earth's surface, $g = 9.8 \text{ m/s}^2$
Boltzmann's constant, $k_B = 1.38 \times 10^{-23} \text{ J/K}$	
1 unified atomic mass unit, $1 \text{ u} = 1.66 \times 10^{-27} \text{ kg} = 931 \text{ MeV}/c^2$	
Planck's constant, $h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s} = 4.14 \times 10^{-15} \text{ eV}\cdot\text{s}$	
	$hc = 1.99 \times 10^{-25} \text{ J}\cdot\text{m} = 1.24 \times 10^3 \text{ eV}\cdot\text{nm}$
Vacuum permittivity, $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$	
Coulomb's law constant, $k = 1/(4\pi\epsilon_0) = 9.0 \times 10^9 (\text{N}\cdot\text{m}^2)/\text{C}^2$	
Vacuum permeability, $\mu_0 = 4\pi \times 10^{-7} (\text{T}\cdot\text{m})/\text{A}$	
Magnetic constant, $k' = \mu_0/(4\pi) = 1 \times 10^{-7} (\text{T}\cdot\text{m})/\text{A}$	
1 atmosphere pressure, $1 \text{ atm} = 1.0 \times 10^5 \text{ N/m}^2 = 1.0 \times 10^5 \text{ Pa}$	

UNIT SYMBOLS	meter, m	mole, mol	watt, W	farad, F
	kilogram, kg	hertz, Hz	coulomb, C	tesla, T
	second, s	newton, N	volt, V	degree Celsius, °C
	ampere, A	pascal, Pa	ohm, Ω	electron volt, eV
	kelvin, K	joule, J	henry, H	

PREFIXES		
Factor	Prefix	Symbol
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

VALUES OF TRIGONOMETRIC FUNCTIONS FOR COMMON ANGLES							
θ	0°	30°	37°	45°	53°	60°	90°
$\sin \theta$	0	$1/2$	$3/5$	$\sqrt{2}/2$	$4/5$	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	$4/5$	$\sqrt{2}/2$	$3/5$	$1/2$	0
$\tan \theta$	0	$\sqrt{3}/3$	$3/4$	1	$4/3$	$\sqrt{3}$	∞

The following assumptions are used in this exam.

- The frame of reference of any problem is inertial unless otherwise stated.
- The direction of current is the direction in which positive charges would drift.
- The electric potential is zero at an infinite distance from an isolated point charge.
- All batteries and meters are ideal unless otherwise stated.
- Edge effects for the electric field of a parallel plate capacitor are negligible unless otherwise stated.



ADVANCED PLACEMENT PHYSICS C EQUATIONS

MECHANICS

$$v_x = v_{x0} + a_x t$$

$$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$$

$$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$$

$$\vec{a} = \frac{\sum \vec{F}}{m} = \frac{\vec{F}_{net}}{m}$$

$$\vec{F} = \frac{d\vec{p}}{dt}$$

$$\vec{J} = \int \vec{F} dt = \Delta \vec{p}$$

$$\vec{p} = m\vec{v}$$

$$|\vec{F}_f| \leq \mu |\vec{F}_N|$$

$$\Delta E = W = \int \vec{F} \cdot d\vec{r}$$

$$K = \frac{1}{2} mv^2$$

$$P = \frac{dE}{dt}$$

$$P = \vec{F} \cdot \vec{v}$$

$$\Delta U_g = mg\Delta h$$

$$a_c = \frac{v^2}{r} = \omega^2 r$$

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\vec{\alpha} = \frac{\sum \vec{\tau}}{I} = \frac{\vec{\tau}_{net}}{I}$$

$$I = \int r^2 dm = \sum mr^2$$

$$x_{cm} = \frac{\sum m_i x_i}{\sum m_i}$$

$$v = r\omega$$

$$\vec{L} = \vec{r} \times \vec{p} = I\vec{\omega}$$

$$K = \frac{1}{2} I\omega^2$$

$$\omega = \omega_0 + \alpha t$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$$

a = acceleration
 E = energy
 F = force
 f = frequency
 h = height
 I = rotational inertia
 J = impulse
 K = kinetic energy
 k = spring constant
 ℓ = length
 L = angular momentum
 m = mass
 P = power
 p = momentum
 r = radius or distance
 T = period
 t = time
 U = potential energy
 v = velocity or speed
 W = work done on a system
 x = position
 μ = coefficient of friction
 θ = angle
 τ = torque
 ω = angular speed
 α = angular acceleration
 ϕ = phase angle

$$\vec{F}_s = -k\Delta x$$

$$U_s = \frac{1}{2} k(\Delta x)^2$$

$$x = x_{\max} \cos(\omega t + \phi)$$

$$T = \frac{2\pi}{\omega} = \frac{1}{f}$$

$$T_s = 2\pi \sqrt{\frac{m}{k}}$$

$$T_p = 2\pi \sqrt{\frac{\ell}{g}}$$

$$|\vec{F}_G| = \frac{Gm_1 m_2}{r^2}$$

$$U_G = -\frac{Gm_1 m_2}{r}$$

ELECTRICITY AND MAGNETISM

$$|\vec{F}_E| = \frac{1}{4\pi\epsilon_0} \left| \frac{q_1 q_2}{r^2} \right|$$

$$\vec{E} = \frac{\vec{F}_E}{q}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

$$E_x = -\frac{dV}{dx}$$

$$\Delta V = -\int \vec{E} \cdot d\vec{r}$$

$$V = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i}$$

$$U_E = qV = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

$$\Delta V = \frac{Q}{C}$$

$$C = \frac{\kappa \epsilon_0 A}{d}$$

$$C_p = \sum_i C_i$$

$$\frac{1}{C_s} = \sum_i \frac{1}{C_i}$$

$$I = \frac{dQ}{dt}$$

$$U_C = \frac{1}{2} Q\Delta V = \frac{1}{2} C(\Delta V)^2$$

$$R = \frac{\rho \ell}{A}$$

$$\vec{E} = \rho \vec{J}$$

$$I = Nev_d A$$

$$I = \frac{\Delta V}{R}$$

$$R_s = \sum_i R_i$$

$$\frac{1}{R_p} = \sum_i \frac{1}{R_i}$$

$$P = I\Delta V$$

A = area
 B = magnetic field
 C = capacitance
 d = distance
 E = electric field
 $\mathcal{E}$ = emf
 F = force
 I = current
 J = current density
 L = inductance
 ℓ = length
 n = number of loops of wire per unit length
 N = number of charge carriers per unit volume
 P = power
 Q = charge
 q = point charge
 R = resistance
 r = radius or distance
 t = time
 U = potential or stored energy
 V = electric potential
 v = velocity or speed
 ρ = resistivity
 Φ = flux
 κ = dielectric constant

$$\vec{F}_M = q\vec{v} \times \vec{B}$$

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I$$

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{\ell} \times \hat{r}}{r^2}$$

$$\vec{F} = \int I d\vec{\ell} \times \vec{B}$$

$$B_s = \mu_0 nI$$

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

$$\mathcal{E} = \oint \vec{E} \cdot d\vec{\ell} = -\frac{d\Phi_B}{dt}$$

$$\mathcal{E} = -L \frac{dI}{dt}$$

$$U_L = \frac{1}{2} LI^2$$





ADVANCED PLACEMENT PHYSICS C EQUATIONS

GEOMETRY AND TRIGONOMETRY

Rectangle

$$A = bh$$

Triangle

$$A = \frac{1}{2}bh$$

Circle

$$A = \pi r^2$$

$$C = 2\pi r$$

$$s = r\theta$$

Rectangular Solid

$$V = \ell wh$$

Cylinder

$$V = \pi r^2 \ell$$

$$S = 2\pi r \ell + 2\pi r^2$$

Sphere

$$V = \frac{4}{3}\pi r^3$$

$$S = 4\pi r^2$$

Right Triangle

$$a^2 + b^2 = c^2$$

$$\sin \theta = \frac{a}{c}$$

$$\cos \theta = \frac{b}{c}$$

$$\tan \theta = \frac{a}{b}$$

A = area

C = circumference

V = volume

S = surface area

b = base

h = height

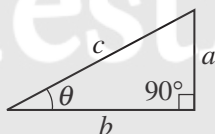
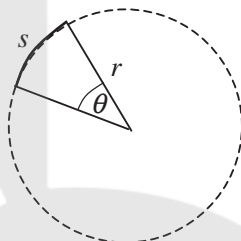
ℓ = length

w = width

r = radius

s = arc length

θ = angle



CALCULUS

$$\frac{df}{dx} = \frac{df}{du} \frac{du}{dx}$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^{ax}) = ae^{ax}$$

$$\frac{d}{dx}(\ln ax) = \frac{1}{x}$$

$$\frac{d}{dx}[\sin(ax)] = a \cos(ax)$$

$$\frac{d}{dx}[\cos(ax)] = -a \sin(ax)$$

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}$$

$$\int \frac{dx}{x+a} = \ln|x+a|$$

$$\int \cos(ax) dx = \frac{1}{a} \sin(ax)$$

$$\int \sin(ax) dx = -\frac{1}{a} \cos(ax)$$

VECTOR PRODUCTS

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$|\vec{A} \times \vec{B}| = AB \sin \theta$$



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Begin your response to **QUESTION 1** on this page.

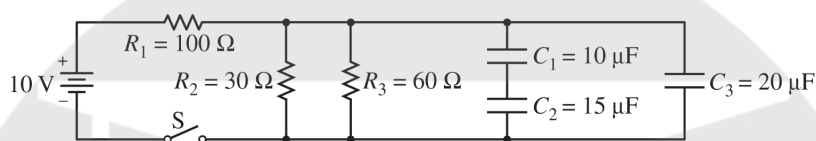
PHYSICS C: ELECTRICITY AND MAGNETISM

SECTION II

Time—45 minutes

3 Questions

Directions: Answer all three questions. The suggested time is about 15 minutes for answering each of the questions, which are worth 15 points each. The parts within a question may not have equal weight. Show all your work in this booklet in the spaces provided after each part.



1. The circuit shown above is composed of an ideal 10 V battery, three resistors and three capacitors with the values shown, and an open switch S. The capacitors are initially uncharged. Switch S is now closed.

(a) Calculate the current through R_1 immediately after switch S is closed.

Switch S has been closed for a long time, and the circuit has reached a steady state.

(b) Calculate the potential difference across R_1 .

(c)

i. Calculate the charge stored on the positive plate of capacitor C_2 .

ii. Is the charge stored on capacitor C_3 greater than, less than, or equal to the charge stored on capacitor C_2 ?

_____ Greater than _____ Less than _____ Equal to

Justify your answer.





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Continue your response to **QUESTION 1** on this page.

Switch S is then opened.

(d)

- Determine the current through R_1 immediately after the switch is opened.
- Calculate the current through R_2 immediately after the switch is opened.

(e) On the axes below, sketch a graph of the potential difference V across capacitor C_2 as a function of time t if switch S is opened at time $t = 0$. Label the maximum value.



Capacitor C_3 is replaced by two $10\ \mu\text{F}$ capacitors connected in series, switch S is closed, and the circuit reaches equilibrium. Switch S is then opened at time $t = 0$.

(f) For $t > 0$, would the sketch of a graph of the new voltage across C_2 as a function of time be above, below, or the same as the sketch for part (e) ?

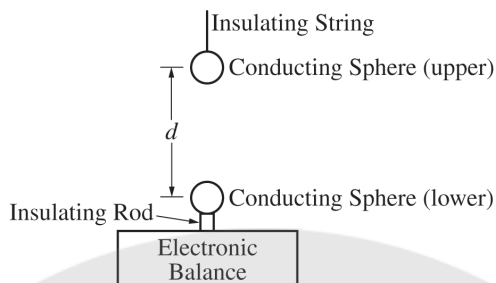
_____ Above _____ Below _____ The same

Justify your answer.



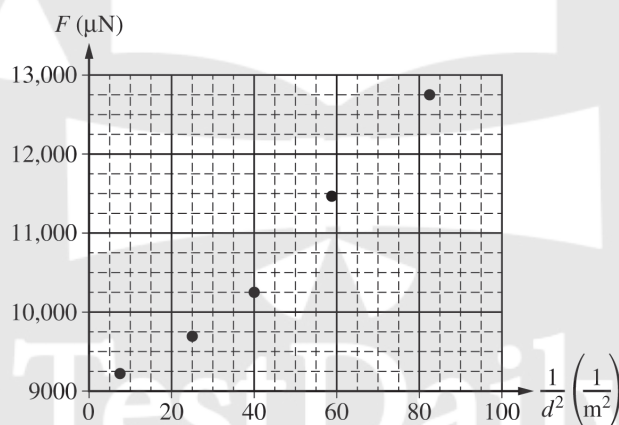
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Begin your response to **QUESTION 2** on this page.



2. Students perform an experiment to study the force between two charged objects using the apparatus shown above, which contains two identical conducting spheres. The upper sphere is attached to an insulating string, which can be used to move the sphere downward. The lower sphere sits on an insulating rod, which is on an electronic balance. The electronic balance is zeroed before the lower sphere and insulating rod are in place.

For the first trial, a charge of Q is placed on each sphere and then the upper sphere is slowly moved downward. The students measure the distance d between the centers of the spheres and the magnitude F of the force that appears on the electronic balance. The recorded data are shown on the graph of F as a function of $\frac{1}{d^2}$ shown below.



(a)

- Draw a line that represents the best fit to the points shown.
- Use the graph to calculate the charge Q .





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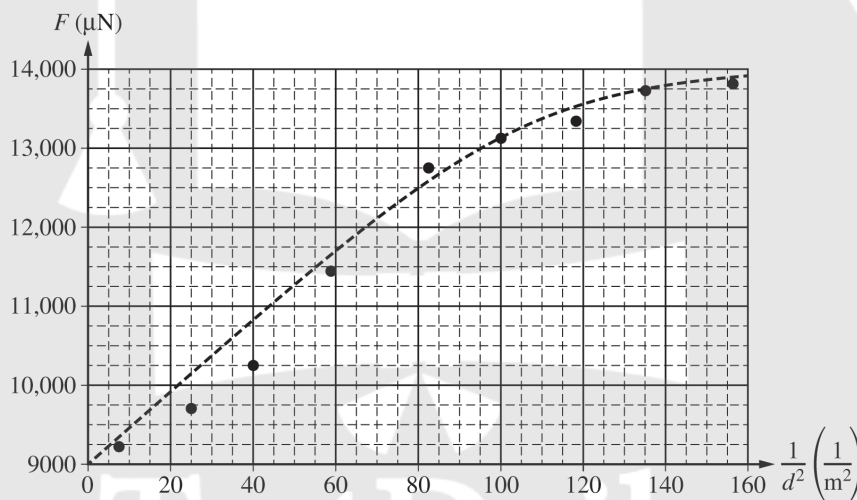
Continue your response to **QUESTION 2** on this page.

iii. On the graph on the previous page, draw a circle around the data point that was taken when the distance between the centers of the spheres was the least.

iv. Determine the distance between the centers of the spheres for the data point indicated above.

v. What physical quantity does the vertical intercept represent?

Justify your answer.



The experiment is extended by collecting additional data points, which appear on the right side of the graph shown above. The new data points do not follow the linear pattern seen with the first points. The group of students tries to explain this discrepancy.

(b) One student suspects that charge is slowly leaking off the top sphere. Could this explain the discrepancy?

____ Yes ____ No

Justify your answer.



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Continue your response to **QUESTION 2** on this page.

(c) A second student suspects that the excess charges have rearranged themselves, polarizing the spheres.

i. On the circles representing the spheres below, use a single “+” sign on each sphere to represent the locations of highest concentration of the excess positive charges.



ii. Explain how this rearrangement could be responsible for the discrepancy.

(d) A third student suggests that the experiment be modified so that the top sphere is given a negative charge that is equal in magnitude to the positive charge given to the bottom sphere.

i. On the circles representing the spheres below, use a single “+” sign on the bottom sphere to represent the location of highest concentration of the excess positive charges. Use a single “−” sign on the top sphere to represent the location of the highest concentration of the excess negative charges.



ii. For a separation distance equal to that of the data point indicated in part (a)(iii), would the magnitude of the force reading with spheres of opposite charges be greater than, less than, or equal to the magnitude of the force reading with spheres of the same charges?

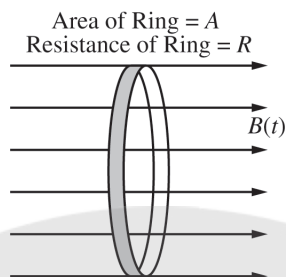
_____ Greater than _____ Less than _____ Equal to

Justify your answer.





Begin your response to **QUESTION 3** on this page.



3. A thin, conducting ring of area A and resistance R is aligned in a uniform magnetic field directed to the right and perpendicular to the plane of the ring, as shown. At time $t = 0$, the magnitude of the magnetic field is B_0 . At $t = 1$ s, the magnitude of the magnetic field begins to decrease according to the equation $B(t) = \frac{\beta}{t}$, where β has units of $\text{T} \cdot \text{s}$.

(a) Derive an equation for the magnitude of the induced current I in the ring as a function of t for $t > 1$ s. Express your answer in terms of β , A , R , t , and physical constants, as appropriate.

Assume $A = 0.50 \text{ m}^2$, $R = 2.0 \Omega$, and $\beta = 0.50 \text{ T} \cdot \text{s}$.

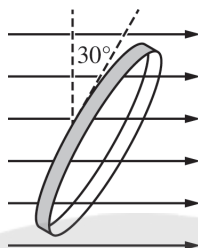
(b) Calculate the electrical energy dissipated in the ring from $t = 1$ s to $t = 2$ s.

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Continue your response to **QUESTION 3** on this page.



The ring is then rotated so that the plane of the ring is aligned at a 30° angle to the magnetic field, as shown.

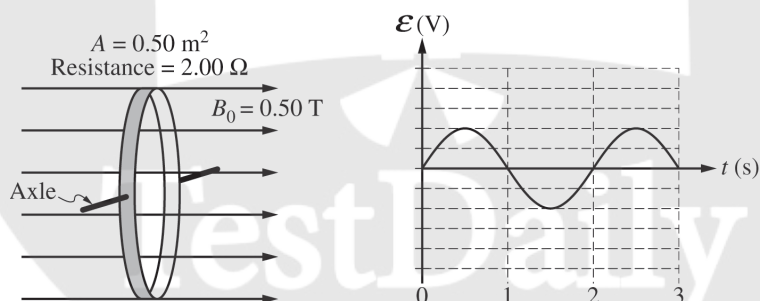
The magnitude of the magnetic field is reset to a magnitude of B_0 at a new time $t = 0$ and again begins to

decrease at $t = 1$ s according to the equation $B(t) = \frac{\beta}{t}$, where β has units of T·s.

(c) Will the amount of energy dissipated in the ring from $t = 1$ s to $t = 2$ s be greater than, less than, or equal to the energy dissipated in part (b) ?

_____ Greater than _____ Less than _____ Equal to

Justify your answer.



The ring is now mounted on an axle that is perpendicular to the magnetic field. The magnitude of the magnetic field is now held at a constant $B_0 = 0.50$ T, as shown. The ring rotates about the axle, and the emf $\mathcal{E}$ induced in the ring as a function of time t is shown on the graph.

(d) Calculate the angular speed ω of the rotating ring in rad/s.





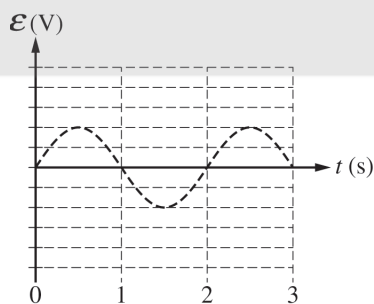
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Continue your response to **QUESTION 3** on this page.

- (e) Calculate the magnitude of the maximum emf $\mathcal{E}_{\text{MAX}}$ induced in the ring.

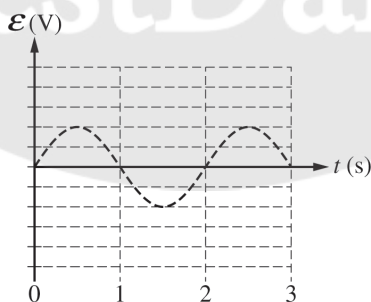
The ring now begins to rotate at an angular speed 2ω .

- (f) On the graph below, draw a curve to indicate the new induced emf $\mathcal{E}$ in the ring. The dashed curve shows the emf induced under the original conditions.



Justify your sketch, specifically identifying and addressing any similarities or differences between the sketch and the original graph.

PRACTICE GRAPH - Use the graph below to practice your sketch for part (f). Any work shown on the graph below will NOT be graded.





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Question 1: Free-Response Question

15 points

- (a) For an indication that the capacitors act like a short circuit immediately after the switch is closed **1 point**

$$R_{EQ} = R_1 = 100 \, \Omega$$

For using Ohm's law to calculate the current through R_1 **1 point**

$$I = \frac{\Delta V}{R} = \frac{(10 \, \text{V})}{(100 \, \Omega)}$$

$$I = 0.10 \, \text{A}$$

Total for part (a) 2 points

- (b) For correctly determining the equivalent resistance during steady-state **1 point**

$$\frac{1}{R_P} = \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{30 \, \Omega} + \frac{1}{60 \, \Omega} \therefore R_P = 20 \, \Omega$$

$$R_{EQ} = R_1 + R_P = 100 \, \Omega + 20 \, \Omega = 120 \, \Omega$$

For correctly calculating the current through the battery **1 point**

$$I = \frac{\Delta V}{R} = \frac{(10 \, \text{V})}{(120 \, \Omega)}$$

$$I = 0.083 \, \text{A}$$

The potential difference across R_1

$$\Delta V = (0.083 \, \text{A})(100 \, \Omega) = 8.3 \, \text{V}$$

Total for part (b) 2 points

- (c) i. For using a correct equation to calculate the potential difference across C_3 **1 point**

$$\Delta V_{C_3} = \Delta V_P = IR_P = (0.083 \, \text{A})(20 \, \Omega) = 1.67 \, \text{V}$$

For using the correct equivalent capacitance of the series combination to calculate the charge stored in capacitor C_2 **1 point**

$$\frac{1}{C_S} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{10 \, \mu\text{F}} + \frac{1}{15 \, \mu\text{F}} \therefore C_S = 6.0 \, \mu\text{F}$$

$$Q_2 = Q_S = C_S \Delta V_{C_{12}} = (1.67 \, \text{V})(6.0 \, \mu\text{F}) = 10 \, \mu\text{C}$$

- ii. For selecting "Greater than" and attempting a relevant justification **1 point**

For a correct justification **1 point**

Example response for part (c)(ii)

The potential difference across the series combination of C_1 and C_2 is the same as the potential difference across C_3 ; thus, capacitor C_3 has a greater potential difference across its plates than C_2 and has a greater capacitance. Because $Q = C\Delta V$, C_3 stores a greater charge.

Scoring note: A justification showing a mathematical proof earns full credit.

Total for part (c) 4 points





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(d) i. For indicating that the current through R_1 is zero when the switch is opened **1 point**

$$I_1 = 0$$

ii. For indicating that the potential difference across R_2 is equal to the potential difference across the capacitors before the switch was opened **1 point**

$$\Delta V_{R_2} = \Delta V_{C_{12}} = 1.67 \text{ V}$$

For using Ohm's law to calculate the current through R_2 **1 point**

$$I = \frac{\Delta V_{R_2}}{R_2} = \frac{(1.67 \text{ V})}{(30 \, \Omega)}$$

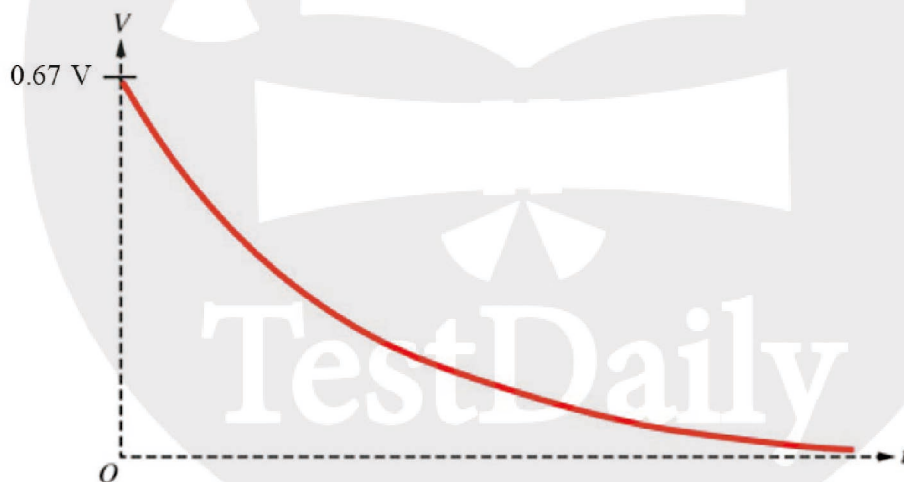
$$I = 0.056 \text{ A}$$

Total for part (d) 3 points

(e) For a curve starting at a nonzero, labeled maximum value **1 point**

For a concave-up curve with the horizontal axis as an asymptote **1 point**

Example response for part (e)



Total for part (e) 2 points

(f) For selecting “Below” and attempting a relevant justification **1 point**

For a correct justification **1 point**

Example response for part (f)

Because the equivalent capacitance of the circuit would decrease, the time constant would also decrease. Thus the capacitors would discharge more rapidly, and the new curve would be below the original curve.

Total for part (f) 2 points

Total for question 1 15 points



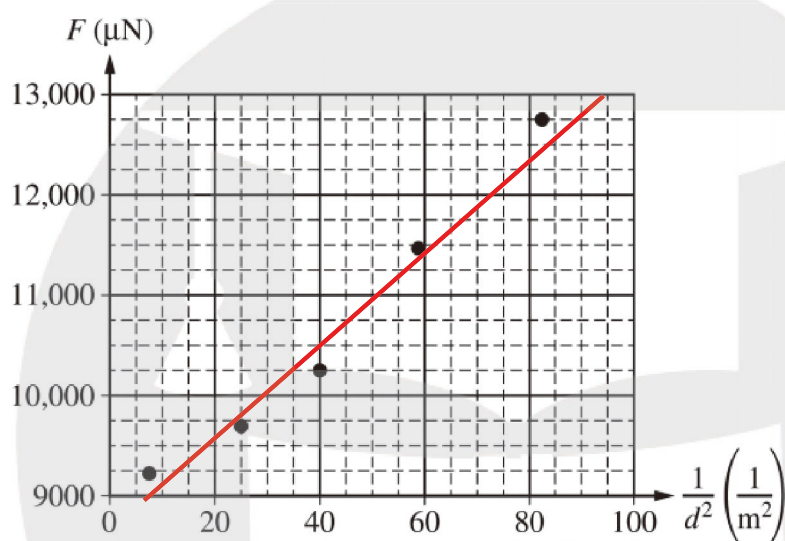
Question 2: Free-Response Question

15 points

- (a) i. For drawing an appropriate best-fit line

1 point

Example response for part (a)(i)



- ii. For calculating the slope using two points from the best-fit line

1 point

$$\text{slope} = m = \frac{\Delta y}{\Delta x} = \frac{(12,000 - 10,500) \mu\text{N}}{(72 - 40) \text{ m}^{-2}} = 4.7 \times 10^{-5} \text{ N} \cdot \text{m}^2$$

For correctly relating the slope to the charge

1 point

$$F = \frac{kQ^2}{r^2} \therefore \text{slope} = kQ^2$$

For correctly calculating the charge consistent with the slope from the best-fit line

1 point

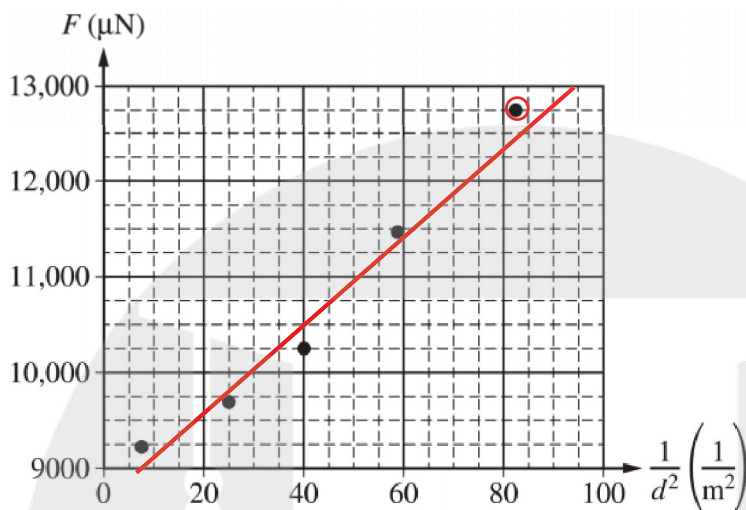
$$Q = \sqrt{\frac{\text{slope}}{k}} = \sqrt{\frac{(4.7 \times 10^{-5} \text{ N} \cdot \text{m}^2)}{(9 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)}} = 7.2 \times 10^{-8} \text{ C}$$





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- (a) iii. For drawing a circle around the appropriate data point **1 point**



- iv. For a correct answer **1 point**

From the graph, $\frac{1}{d^2} = 82.5 \text{ m}^{-2}$

$$d = 0.11 \text{ m}$$

- v. For indicating that the y-intercept should be the weight of the conducting sphere and insulating rod **1 point**

For a correct justification **1 point**

Example responses for part (a)(v).

When the spheres are infinitely far apart, there is no electric force between them. The only force recorded by the balance must be the weight of the lower sphere and insulating rod.

OR

The scale is zeroed before the insulating rod, and lower sphere are placed on it, so the nonzero y-intercept is their weight.

Total for part (a) 8 points

- (b) For selecting “Yes” and providing a correct justification **1 point**

Example response for part (b)

Because the later force measurements are lower than the expected values on the best-fit line, then a decrease in charge could explain the discrepancy seen in the data.



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(c) i.	For correctly indicating the position of the excess positive charges	1 point
	For correctly indicating that the net charge is positive	1 point

Example response for part (c)(i)



ii.	For a correct justification	1 point
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Example response for part (c)(ii)

On a conductor, excess charges are free to move and will repel each other. The excess charges in the sphere will rearrange themselves until they are as far apart from each other as possible; thus, the repelling force decreases.

Total for part (c) 3 points

(d) i.	For appropriately indicating the position of the excess charges on both spheres	1 point
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ii.	For selecting “Less than” and attempting a relevant justification	1 point
	For a correct justification	1 point

Example response for part (d)

Because the opposite charged spheres will attract each other, this will decrease the upward normal force exerted by the scale on the lower sphere; thus, the reading on the scale will be less than when the spheres had the same charge.

Total for part (d) 3 points

Total for question 2 15 points





Question 3: Free-Response Question

15 points

(a) For using Faraday’s law to relate emf to change in flux 1 point

$$\mathcal{E} = \left| \frac{d\Phi}{dt} \right| = \left| \frac{d(B \cdot A)}{dt} \right|$$

For a correct expression for the emf 1 point

$$\mathcal{E} = \left| A \frac{dB}{dt} \right| = \left| A \frac{d\left(\frac{\beta}{t}\right)}{dt} \right| = \left| -\frac{A\beta}{t^2} \right| = \frac{A\beta}{t^2}$$

For correctly substituting into Ohm’s law 1 point

$$I = \frac{\Delta V}{R} = \frac{\mathcal{E}}{R} = \frac{A\beta}{Rt^2}$$

Scoring note: Full credit is earned if the negative sign is included.

Total for part (a) 3 points

(b) For using a correct equation to calculate the energy dissipated in the ring 1 point

$$E = \int P dt = \int I^2 R dt$$

For correctly substituting into the equation above 1 point

$$E = \int \left(\frac{A\beta}{Rt^2} \right)^2 R dt = \int \frac{A^2 \beta^2}{R^2 t^4} R dt$$

For indicating the correct limits or constant of integration 1 point

$$E = \frac{A^2 \beta^2}{R} \int_1^2 \frac{1}{t^4} dt$$

For a correct answer with units 1 point

$$E = \frac{A^2 \beta^2}{R} \left[-\frac{1}{3t^3} \right]_1^2 = \frac{(0.50 \text{ m}^2)^2 (0.50 \text{ T}\cdot\text{s})^2}{(3)(2.00 \text{ }\Omega)} \left(-\frac{1}{2^3} - \left(-\frac{1}{1^3} \right) \right) = 0.0091 \text{ J}$$

Total for part (b) 4 points

(c) For selecting “Less than” and attempting a relevant justification 1 point

For a correct justification 1 point

Example response for part (c)

After the ring rotates, fewer magnetic field lines pass through the ring, and the flux is less than before. Thus, the emf, current, and rate at which energy is dissipated is less.

Total for part (c) 2 points

(d) For using a correct equation to calculate angular speed 1 point

$$\omega = \frac{2\pi}{T}$$



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For substituting appropriate values from the graph in the correct equation

1 point

$$\omega = \frac{(2\pi)}{(2 \text{ s})} = 3.14 \frac{\text{rad}}{\text{s}}$$

Total for part (d) 2 points

(e) For using a correct equation for solving for the emf in terms of the angular speed

1 point

$$\mathcal{E} = \frac{d\Phi}{dt} = \frac{d(B \cdot A)}{dt} = BA \frac{d}{dt}(\cos(\omega t)) = -BA\omega \sin(\omega t)$$

For correctly substituting into the correct equation

1 point

$$\mathcal{E}_{\text{MAX}} = BA\omega = (0.50 \text{ T})(0.50 \text{ m}^2)\left(3.14 \frac{\text{rad}}{\text{s}}\right) = 0.79 \text{ V}$$

Total for part (e) 2 points

(f) For a sine curve with half the period of the dashed curve and a correct justification

1 point

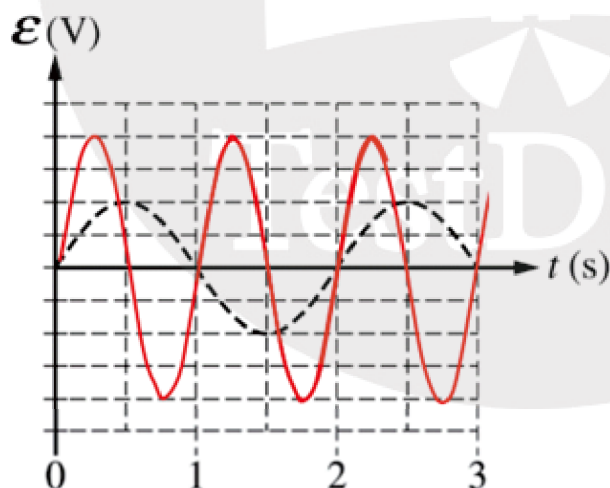
For a sine curve with twice the amplitude of the dashed curve and a correct justification

1 point

OR

a sine curve with an amplitude consistent with work shown in part (e)

Example response for part (f)



Because the rotational speed is twice the original speed, then, according to the equation $|\mathcal{E}| = B_0 A \omega \sin(\omega t)$ this will double the amplitude of the curve and cut the period in half.

Total for part (f) 2 points

Total for question 3 15 points



